

PETER KIM

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EDUCATION

University of Michigan College of Engineering

Bachelor of Engineering in Electrical Engineering

Ann Arbor, MI

Expected Graduation: May 2029

Coursework: EECS 281 (Data Structures & Algorithms), EECS 280 (Programming & Data Structures), EECS 215* (Circuits), EECS 270 (Digital Logic), ROB 380* (Autonomous Robotics), EECS 230* (Electromagnetics I), MATH 214 (Applied Linear Algebra), EECS 203 (Discrete Math) (*in progress)

Wayne State University

Completed coursework toward Bachelor of Engineering in Electrical Engineering

Detroit, MI

Aug 2024 – May 2025

Research: Implemented and evaluated PyTorch denoising pipelines for AI diffusion models under faculty research in probabilistic generative models

EXPERIENCE

Software Engineering Co-Op / Intern

Yazaki North America

Aug 2025 – Present; May 2025 – Aug 2025

Canton, MI

- Built a wire-harness schematic QA pipeline (Python, pdfplumber/Tesseract OCR + validation rules) that auto-locates tapping and layer defects in production drawings and exports annotated review bundles; root-caused a systematic defect driving ~\$44K savings per frame harness and cut review time 88% (128 hrs → <16 hrs)
- Owned an internal Python service end to end — design, deployment, documentation, and ongoing user support — extracting Assembly BOMs from Tesla EV wiring-harness data via format-specific parsers with a vision-model OCR fallback; validated 17/17 parts on a production harness and 44/50 (88%) on image-only drawings against a 0-line incumbent baseline

Creative Video Intern

University of Michigan Athletics

May 2026 – Present

Ann Arbor, MI

- Built *um-autolog*, an on-device detection and localization pipeline replacing hours of manual video logging; person detection localizes subjects in frame, character recognition is constrained to per-subject crops to suppress false positives, and candidate identifications are ranked by weighted voting across frames; documented and covered by 33 unit tests

PROJECTS

Autonomous Gimbal Robot (In Progress) | ROS 2 Jazzy, Raspberry Pi 5, Embedded Linux, LiDAR, C++, Python

- Brought up an embedded Linux compute stack on Raspberry Pi 5 (Ubuntu 24.04, ROS 2 Jazzy) and integrated dual MIPI CSI cameras and a 2D LiDAR, writing ROS 2 nodes in C++/Python directly against vendor drivers and hardware datasheets
- Designed the electromechanical stack for the robot: sized and regulated a LiPo power system (load budget, rail regulation, current headroom) and selected motor drivers off datasheet stall current for a holonomic drivetrain

Autonomous Snowplow Robot (Team SnowBlo) | Arduino, Raspberry Pi, C/C++, Python, OpenCV, MQTT

- Wrote the Arduino motor-control firmware in C/C++ and integrated it with a Raspberry Pi perception stack and a custom 3D-printed chassis into one working system with a 4-person team
- Implemented the detection and pose-estimation stack: AprilTag detection with OpenCV homography to project the camera view into a bird's-eye frame, feeding a differential-drive navigation controller; calibrated camera intrinsics and steering parameters against field measurements
- Debugged hardware and software together to close the loop from sensing to actuation, streaming real-time telemetry over UART/serial and MQTT for live bring-up and tuning

TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB, Java, JavaScript

Software: Linux (Ubuntu, embedded), ROS 2 Jazzy, Git, GitHub Actions (CI/CD), unit testing, technical documentation, PyTorch

Detection & Imaging: OpenCV, camera calibration and intrinsics, homography and coordinate transforms, AprilTag detection, multi-frame estimation

Hardware & Instrumentation: Raspberry Pi 5, Arduino, MCU firmware, MIPI CSI cameras, 2D LiDAR, motor drivers, LiPo power, UART/serial, MQTT, Inventor CAD, 3D printing